

Predicting Used Toyota Corolla Prices

Model development, diagnostics, and the limits of a causal reading

1,429 cars · 45 predictors · 19 specifications · one holdout, opened once

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Data Analysis with Statistical Software · Dr. Orit Rafaeli · Final project

Two questions, not one

1 How accurately can we predict a Corolla's price?

Criterion: out-of-sample RMSE in euros.

Variables selected by cross-validation and regularisation — never by filtering p-values.

2 What is one feature actually worth?

Criterion: absence of bias.

p-values return, for a different job — testing whether the two groups of cars are comparable at all.

A model that wins the first question is not entitled to answer the second. Following Shmueli (2010), the two are kept apart from the first cell of the notebook to the last.

What we removed, and the one removal that mattered

Records	Decision	Why	What it cost
5 Verso MPVs	Excluded	7-seat MPV, 1,320–1,480 kg against a Corolla median of 1,067 kg. A population definition, not outlier removal.	2.5 €
CC = 16,000 (Id 81)	Corrected to 1,600	The Model string reads "1.6" — the record contains its own correction.	—
2 impossible weights	Excluded	More than 5 SD from their own specification group ($z = 16.8$ and 11.5). Rule declared before the result was seen.	102 €
2 cars at $KM \leq 1$	Flagged, kept	Implausible, but nothing in the record proves them wrong.	–8.9 €

Two typed rows out of 859 are worth more than every modelling choice in the project. Putting them back costs 102 € of error and collapses the Weight coefficient from 17.36 €/kg to 2.97. The gap between the best and worst estimator here is smaller than that.

The rules were fixed before a single model was fitted

857 / 572

60/40 stratified split

KS p = 0.95, 1.00, 0.99
on Price, Age and KM

3 × 10-fold

One shared CV protocol

Every candidate scored on
identical folds, seed 2026

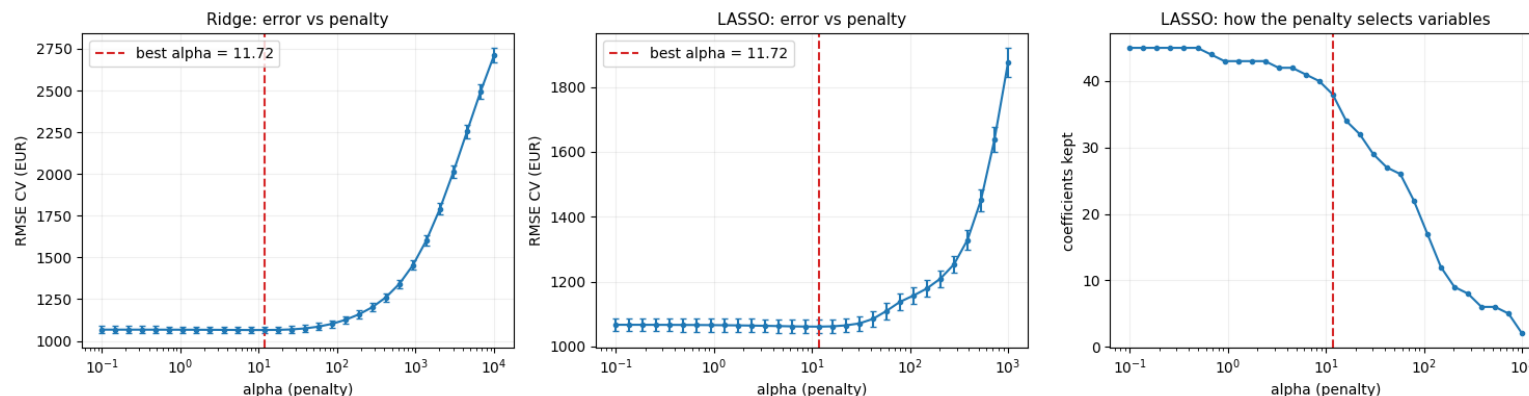
1 × SE

Pre-declared tie-break

Fewer parameters → no log
target → no penalty

- **RMSE in euros is the primary metric.** MAE, MAPE, mean error and R^2 are reported alongside it and never decide.
- **Superiority must be tested, not asserted.** Paired Wilcoxon on per-car absolute errors, Holm-corrected — paired on the car, not the fold.
- **The holdout opens once, after the model is named.** The candidate list is hashed and closed first. A model that wins there is not re-crowned.

Result 1: the penalty buys nothing



Both grids select $\lambda \approx 11.72$. Error is flat across the lower half of each grid — the penalty changes which variables survive long before it changes accuracy.

And it gets blunter on the final design.

Re-run on the 51-column specification the diagnosis produced, both grids bottom out on the boundary at $\lambda = 0.10$ — cross-validation asking for no penalty at all — and LASSO keeps all 51 columns. OLS 976.2, LASSO 978.1, Ridge 980.3 €.

Model	RMSE CV	SE	Kept
OLS	1,067.3 €	20.3	45
Ridge	1,064.9 €	20.6	45
LASSO	1,061.4 €	21.1	38

A 6 € spread against a standard error of 20.

With 857 rows against 45 predictors, OLS is already stable — there is almost no variance left for shrinkage to remove.

What LASSO bought was interpretation, not accuracy

Fuel_Type_Diesel — the worst collinearity offender in the project (VIF 23.5; auxiliary $R^2 = 0.957$ on HP, CC, Weight and Quarterly_Tax)

+25 €

under OLS

kept, positive

-28 €

under Ridge

kept, negative

exactly 0

under LASSO

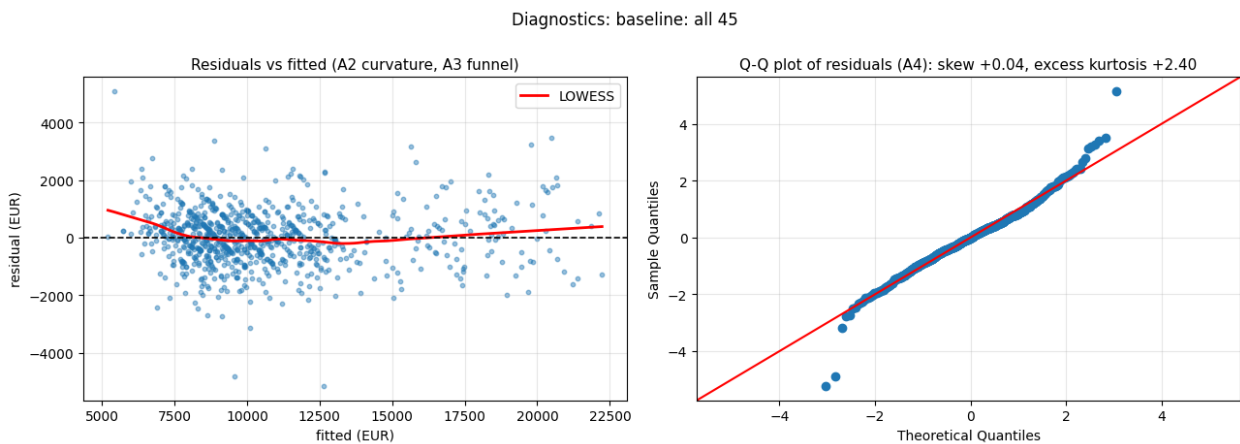
removed

A coefficient whose sign depends on which estimator you use is not measuring anything about diesel engines.

Ridge stabilises the collinear cluster by spreading weight across its members; only LASSO resolves it by removing the redundant column.

Meanwhile the real signals are untouched by either penalty — Age moves $-1,908 \rightarrow -1,804 \rightarrow -1,875$ € per standard deviation. Shrinkage is acting on collinear noise, not on structure.

Five of six assumptions do not hold



	Test	Result	
A1	Durbin–Watson	1.846	pass
A2	Ramsey RESET	F = 40.5	fail
A3	Breusch–Pagan	LM = 234.2	fail
A4	Shapiro–Wilk	kurtosis +2.40	fail
A5	Max VIF	23.5	fail
A6	Cook's D	max 0.144	marginal

The point estimates are usable. The inference around them is not.

The model reaches $R^2 = 0.918$ and satisfies almost none of the conditions under which its standard errors and p-values mean what they claim. Two details set the whole agenda for Part C: normality fails purely through the *tails* (skew +0.04, kurtosis +2.40), which points at a robust loss rather than a target transformation — and the collinearity is one cluster, not ten problems, because diesel Corollas *are* the heavy, large-engine, high-tax cars.

The curvature is between the variables, not inside them

Specification	k	RMSE CV	RESET F (A2)	Max VIF (A5)
Baseline, all 45 predictors	45	1,067.3 €	40.5	23.5
+ 6 forward-selected terms	51	976.2 €	16.3	16,936
+ 17 pure powers instead	62	1,104.2 €	75.2	241.0
Pruned until every VIF < 5	44	1,402.1 €	48.9	4.31

+ 91 €

Six cross-products are the largest single gain anywhere in the project — about four standard errors.

– 37 €

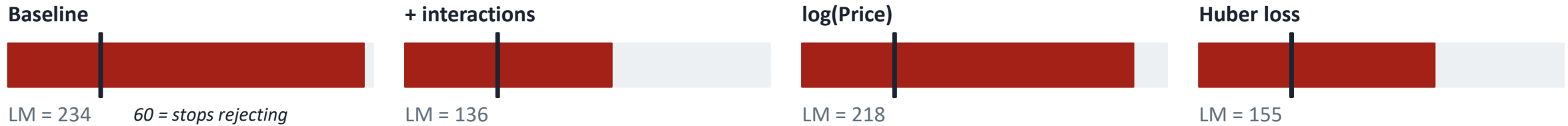
Seventeen pure powers make linearity worse, not better. The bend is not inside any one column.

– 375 €

VIF-pruning is the only thing that makes A5 pass. It deletes Age and Weight themselves.

Result 2: one assumption resisted everything

Breusch–Pagan needs to fall below roughly 60 to stop rejecting. Four unrelated remedies:



Homoscedasticity fails in all 19 runs, without exception. That is the finding, not a failure of the search — a 20,000 € car is genuinely harder to price than a 5,000 € one.

So the remedy matches the diagnosis. Under heteroscedasticity OLS stays unbiased and consistent; what breaks is the variance formula. HC3 estimates each observation's error variance from its own squared residual, weighted by leverage — it changes no coefficient and no prediction, only the uncertainty around them.

1.18×

median SE

inflation under HC3

2.43×

worst case

on log(KM)

4 of 52 coefficients lose significance under HC3. The default output would have reported four findings the data does not support.

The holdout, opened once

Paired Wilcoxon with Holm correction called the top of the table indistinguishable, so the pre-registered one-SE rule decided. Ceiling 991.5 € — five candidates inside it.

Candidate	Params	RMSE CV	RMSE holdout	Optimism	R ²
Forward-selected 6 (45+6) — preferred	52	976.2 €	1,088.6 €	+112	0.900
LASSO on 45+6	52	978.1 €	1,088.8 €	+111	0.900
Ridge on 45+6	52	980.3 €	1,086.4 €	+106	0.900
All 10 candidates (45+10)	56	982.2 €	1,082.1 €	+100	0.900
Huber (45+6)	52	983.1 €	1,086.5 €	+103	0.900
OLS, all 45 (baseline)	46	1,067.3 €	1,123.2 €	+56	0.900

The preferred model comes fourth on the holdout, by 6 € — and is not re-crowned. It was named before the holdout opened; picking the winner afterwards would turn 572 protected cars into a second validation set. Rank correlation between CV and holdout: 0.850.

Optimism is the price of selection. Models that chose their own terms on the folds lose 100–112 €. Models that selected nothing lose 26–56 €.

The 8,622 € that isn't there

Treatment: Automatic_airco — digital climate control, specified at the factory. 76 of 1,429 cars (5.3%).

Specification	Coefficient	HC3 SE
Raw gap, no controls	8,622 €	396
OLS, all 45 base predictors	1,826 €	284
Forward selection (18)	1,975 €	242
OLS, 45+6 — the preferred model	1,210 €	266

Controls absorb 79% of the gap. What is left is not an effect — it moves by 765 € across four defensible specifications, three times either standard error.

What we would collect instead. The factory build sheet and the option's list price; transaction price and days on the lot; regional demand. And above all a source of exogenous variation — a model year in which the package became standard (DiD), dealer-level availability (IV), or a trim threshold (RD).

SMD = -2.09

on age. Treated cars average 24.7 months against 57.9 — the groups barely share a range.

55 of 76

treated cars sit in one trim level (SOL). Only 2 of 896 TERRA cars are treated.

49%

of the 6,797 € bias is age alone — removed by a linear term Part C showed to be the wrong shape.

What the project actually established

Prediction

RMSE 1,088.6 € on 572 unseen cars — about 8% of a typical price, $R^2 = 0.900$. The one line of attack that improved accuracy was the specification: six interaction terms, worth 91 €.

Diagnosis

No specification passes all six assumptions. Linearity responds to treatment; heteroscedasticity does not, and is answered with HC3 rather than another transformation.

Causation

Nothing here licenses it. The remaining 1,210–1,975 € is a partial association over two groups that barely overlap, fitted with a form chosen to minimise prediction error.

For a dealership: price within a ± 820 € band, not to the euro; add back the ~ 100 € the model runs low on unseen cars; expect 120 €/month of depreciation on well-kept cars against 101 € at the median; and audit the stock sheet — two mistyped weights cost more than the gap between every model in this study.

Every assumption, across the specifications that mattered

Run	k	RMSE CV	A2 RESET	A3 BP	A4 kurt.	A5 VIF
Baseline, all 45	45	1,067.3	40.5	234.2	2.40	23.5
√KM (replacing KM)	45	1,080.3	28.9	250.4	2.63	23.5
+ 3 squares + 3 interactions	51	1,026.8	11.3	185.0	1.23	27.5
VIF-pruned to < 5	44	1,402.1	48.9	158.7	1.88	4.31
Pure powers (45+17)	62	1,104.2	75.2	189.4	1.63	241.0
Compact: Age+KM+Weight+HP	4	1,258.4	111.5	73.4	3.08	1.94
All 10 candidates (45+10)	55	982.2	2.47	142.9	0.62	267,969
Forward-selected 6 (45+6)	51	976.2	16.3	136.0	0.60	16,936
log(Price), all 45	45	1,064.9	4.67	217.6	3.24	23.5
Huber (45+6)	51	983.1	0.66	—	—	16,936
Median regression (45+6)	51	1,130.2	0.96	—	—	16,936

19 runs in total. Robust and quantile fits are not graded on A3 or A4 — they were chosen because those conditions do not hold, so scoring them as failures would be a category error.

Does every Corolla depreciate at the same rate?

Quantile regression estimates a conditional percentile of price, not a confidence level. $\tau = 0.1$ is the cheap end of a given specification, $\tau = 0.9$ the best-kept examples.

€ per unit	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	OLS	SE on Age
Age (per month)	-107.1	-101.6	-120.1	-103.6	4.2 – 5.3
KM (per 1,000 km)	-15	-15	-14	-14	—
Weight (per kg)	14.2	17.7	18.2	19.8	—

Only the premium end prices age differently. The budget and median estimates are the same number twice — 5 € apart against standard errors near 4.5. The premium estimate stands 18.5 € per month steeper, about 2.7 standard errors. Mileage and weight differ well inside the noise, so no claim is made about them.

So what: for nine cars in ten the single conditional-mean coefficient is the right summary. At the top of the market it understates monthly depreciation by roughly 16% — a real limitation of the preferred model, at one tail rather than across the market.

Does the conclusion survive the cleaning decisions?

Check	n train	RMSE CV	vs main	Weight coefficient
Main analysis	857	976.2 €	—	17.36 €/kg
Drop the 2 suspected-mileage cars	855	967.4 €	−8.9 €	18.80 €/kg
Re-include the 2 suspected weights	859	1,078.1 €	+101.8 €	2.97 €/kg
Re-include the 5 Verso rows	862	978.7 €	+2.5 €	16.74 €/kg

Two of the three decisions cost almost nothing. Verso costs 2.5 €, which means the exclusion was defensible on its own grounds — a population definition — rather than a convenient improvement. Keeping the two implausible-mileage cars costs under 9 €, a fair price for not deleting data on suspicion.

The weight audit is the one that carries weight. Two records out of 859 nearly erase one of the strongest relationships in the data. The 5-SD rule was declared in §1.3 before the result was inspected, which is exactly what makes this table evidence rather than a rationalisation.